

# L02: The Mathematical Building Blocks of Neural Networks

CSci 560 Deep Learning w/ Python (Chollet) Ch. 2

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Summer 2025

# L02.1 A First Look at a Neural Network

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# L02.2 Data Representations for Neural Networks

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# L02.3 The Gears of Neural Networks: Tensor Operations

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# Matrix · Matrix Dot-Product

- $x$ ,  $y$  are input matrices of shape  $(a, b)$  and  $(b, c)$  respectively
- Each row of  $x$  performs an inner product with corresponding column of  $y$
- So resulting element  $z[i, j]$  is the inner product between row  $i$  of  $x$  and column  $j$  of  $y$
- Resulting shape of  $z$  is  $(a, c)$

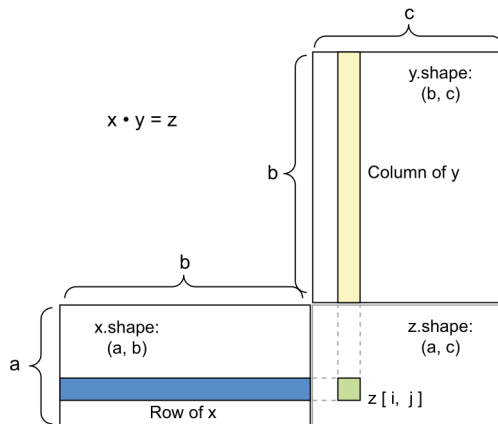


Figure 1: Matrix dot-product box diagram. (Chollet, 2021, pg.43)



# Vector Addition is Translation

- Adding a vector to a point will move the point by a fixed amount.
- Applied to a set of points, result is a translation.

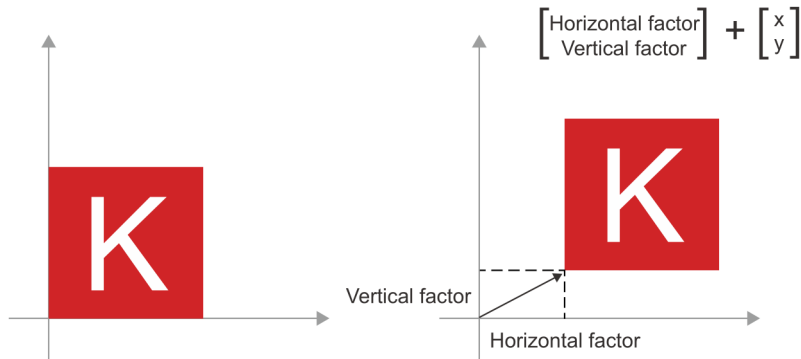


Figure 2: 2D translation as vector addition. (Chollet, 2021, pg.45)

# Matrix Multiplication (dot product) is Rotation

- A counterclockwise rotation by angle  $\theta$  achieved by dot product.

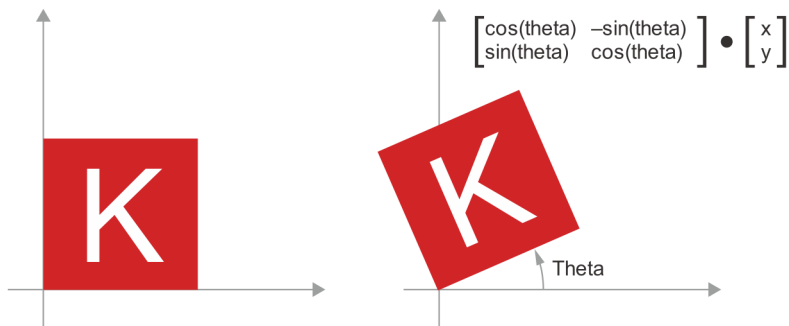


Figure 3: 2D rotation (counterclockwise) as a dot product. (Chollet, 2021, pg.45)

# Matrix Multiplication can also Scale

- Vertical and horizontal scaling also achieved by dot product with a suitable matrix.

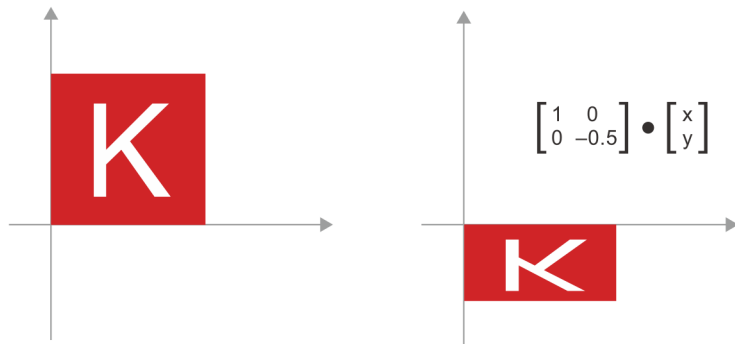


Figure 4: 2D scaling as a dot product. (Chollet, 2021, pg.46)



# Affine Transformation: Linear transform and translation

- Translation, Rotation, Scaling are all linear transformations.
- Can combine, for example  $y = W \cdot x + b$  is a linear transformation plus a translation.

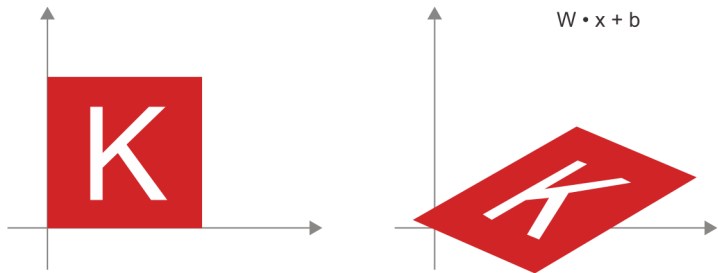


Figure 5: Affine transform in the plane. (Chollet, 2021, pg.46)

# relu Nonlinear Activation Function

- The previous are just combinations of linear transformations.
- Multilayer NN made entirely of linear transformations can be reduced to a single linear transformation.
- Activation functions produce a nonlinear translation.

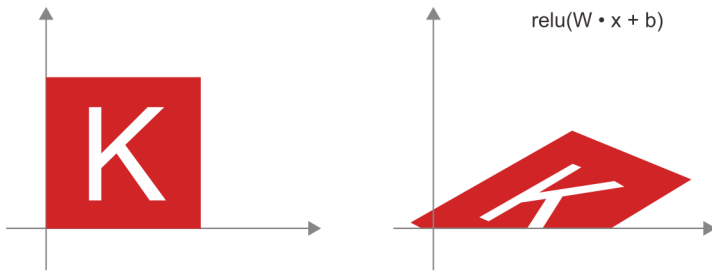


Figure 6: Affine transform followed by (nonlinear) 'relu' activation. (Chollet, 2021, pg.47)

# A Geometric Interpretation of Deep Learning

- NN consist entirely of chains of tensor operations.
- Can be interpreted as geometric transformations of the input data into a new space/shape/manifold.
- Can interpret a NN as a very complex geometric transformation in a high-dimensional space, implemented via a series of simple linear + nonlinear transformations.



Figure 7: Uncrumpling a complicated manifold of data. (Chollet, 2021, pg.47)



# L02.4 The Engine of Neural Networks: Gradient-based Optimization

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# 1D Mini-Batch Stochastic Gradient Descent

- Stochastic because each batch of data is drawn at random.
- Need a reasonable value for the `learning_rate`.
  - too small, slow convergence
  - too large, diverges

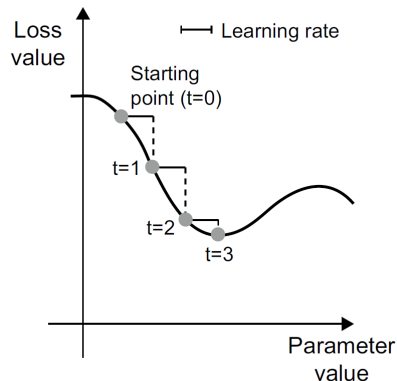
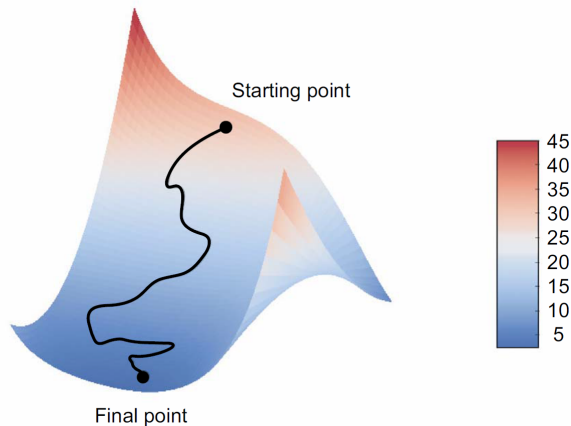


Figure 8: SGD down a 1D loss curve (one learnable parameter) (Chollet, 2021, pg.53)

# 2D Mini-Batch Stochastic Gradient Descent



- Gradient descent will be used in highly dimensional spaces.
- Every weight coefficient in a neural network is a free dimension in the space.
  - Tens of thousands or even millions of parameters.
- Cannot visualize, and intuitions from low-dimension examples can sometimes be misleading.

Figure 9: Gradient descent down a 2D loss surface (two learnable parameters). (Chollet, 2021, pg.54)

# SGD with Momentum / Advanced Optimization Methods

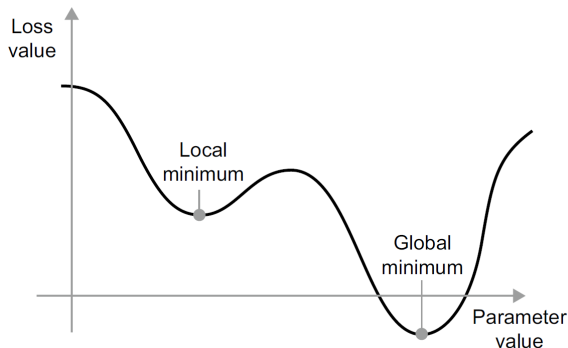


Figure 10: A local minimum and a global minimum.  
(Chollet, 2021, pg.54)

- **Momentum** take into account previous weight updates rather than just looking at current gradient.
- SGD w/ momentum, Adagrad, RMSprop
- Momentum addresses two issues with SGD: convergence speed and local minima.
  - If enough momentum won't get stuck in local minimum
  - Will adjust learning rate to to make sure convergence does not take too long.
- Implemented by moving each step based on both slope (current acceleration) and velocity.

# Chaining Derivatives: The Backpropagation Algorithm

- **Backpropagation** is a way to use the derivatives of simple operations to easily compute the gradient of arbitrarily complex combinations of atomic operations.
- A NN consists of many tensor operations chained together, each of which has a simple known derivative.
- The chain rule states that
$$\text{grad}(y, x) == \text{grad}(y, x1) * \text{grad}(x1, x)$$
- Can be applied to any number of intermediate functions

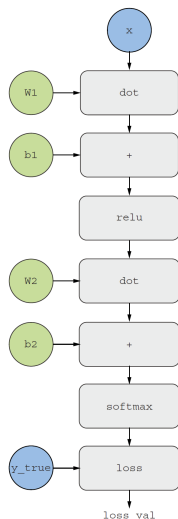
$$\text{grad}(y, x) == (\text{grad}(y, x3) * \text{grad}(x3, x2) * \text{grad}(x2, x1) * \text{grad}(x1, x))$$

```
def fg(x):  
    x1 = g(x)  
    y = f(x1)  
    return y  
  
def fghj(x):  
    x1 = j(x)  
    x2 = h(x1)  
    x3 = g(x2)  
    y = f(x3)  
    return y
```





# Computation Graphs and Automatic Differentiation



- Useful way to think of and perform backpropagation: use computation graphs.
- A **computation graph** is a directed acyclic graph of operations.
- Treat computation as data:
  - Graph can be built in TensorFlow from a GradientTape
  - Represents forward pass
  - Can reverse links to perform backpropagation backward pass.

Figure 11: The computation graph

# Automatic Differentiation: Forward Pass

- Consider a simplified computation graph.
  - single layer all variables are scalar
  - scalar  $w$  for weights and  $b$  for bias
  - scalar  $x$  input
  - no activation function
  - loss is difference of result and  $y_{\text{true}}$
- We want to update  $w$  and  $b$  in a way that will minimize  $\text{loss\_val}$ .
  - want to compute  $\text{grad}(\text{loss\_val}, b)$  and  $\text{grad}(\text{loss\_val}, w)$ .
- Propagate these values to all nodes in the graph from top to bottom, until we reach  $\text{loss\_val}$ .
  - This is the **forward pass**

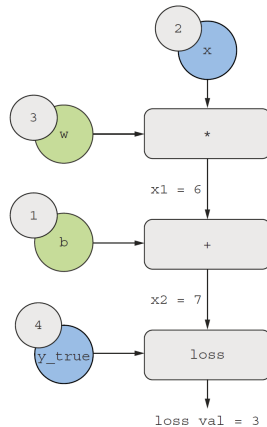


Figure 12: Running a forward pass.  
(Chollet, 2021, pg.58)



# Automatic Differentiation: Backward Pass / Backpropagation

- “reverse” the graph
  - For each edge from A to B, create opposite edge from B to A
  - backward graph represents the **backward pass**, a concrete implementation of performing **backpropagation**
  - You can obtain the derivative of a node with respect to another node by *multiplying the derivatives for each edge along the path linking the two nodes*.

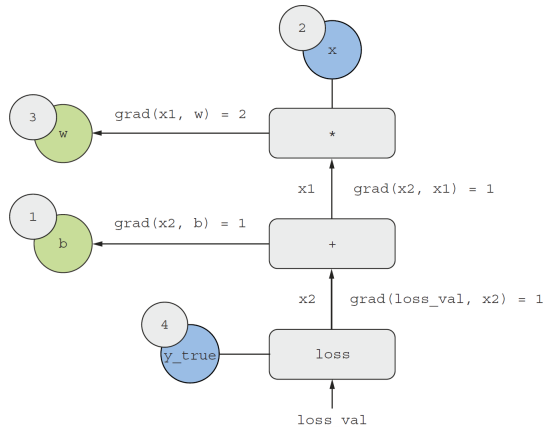


Figure 13: Running a forward pass. (Chollet, 2021, pg.58)



# Automatic Differentiation using the Computation Graph

- Chain rule says the backward graph can be used to obtain the derivatives of the learnable parameters with respect to the loss.
- Multiply the derivatives for each edge along the path linking two nodes.**
- For instance:

$$\begin{aligned} \text{grad}(\text{loss\_val}, w) = & \text{grad}(\text{loss\_val}, x2) * \\ & \text{grad}(x2, x1) * \\ & \text{grad}(x1, w) \end{aligned}$$

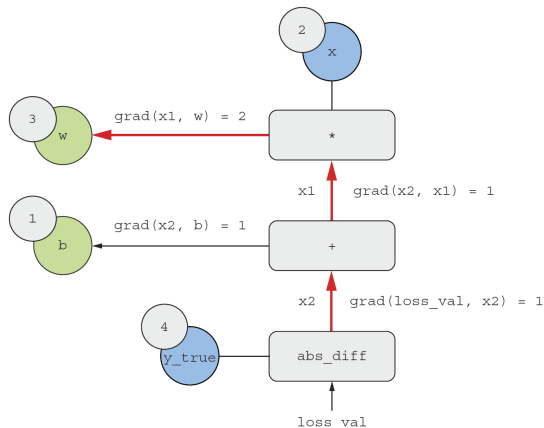


Figure 14: Path from  $\text{loss\_value}$  to  $w$  in the backward graph. (Chollet, 2021, pg.59)



# L02.5 Looking Back at our First Neural Network Example

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Chollet, F. (2021). *Deep learning with python* (second). Manning.